EXPERIMENT 5 : INFORMATION RETRIEVAL USING SpaCy

AIM:

To implement an information retrieval system using spaCy and TF-IDF by preprocessing text, removing stop words, analyzing part-of-speech tags, and retrieving the most relevant documents for a user query using cosine similarity.

ALGORITHM:

1. Import spaCy, Pandas, fetch_20newsgroups, TfidfVectorizer, and cosine_similarity.

2. Load the 20 Newsgroups dataset and select the first 500 documents, excluding headers, footers, and quoted text.

3. Define a preprocessing function to convert text to lowercase, tokenize it with spaCy, remove stop words and non-alphabetic tokens, and use lemmas.

4. Store the original text and preprocessed text in a Pandas DataFrame.

5. Extract stop words from the documents using spaCy and display sample results.

6. Print part-of-speech tags for tokens in the first three documents.

7. Convert the cleaned documents into TF-IDF vectors using TfidfVectorizer.

8. Enter the query “NASA space launch” and preprocess it using the same text-preprocessing function.

9. Transform the query into a TF-IDF vector and calculate cosine similarity between the query and all documents.

10. Sort the similarity scores and display the top five most relevant documents.

Information Retrieval (IR) using spaCy

Step 1: Import Libraries and Load spaCy Model

```
#Information_Retrieval_3
#Shailesh kumar V(241801256)
import spacy
import pandas as pd
import string
from sklearn.datasets import fetch_20newsgroups
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
nlp = spacy.load("en_core_web_sm")
```

Step 2: Load Dataset

```
#Information_Retrieval_3
#SANJAY_S_241801246
newsgroups = fetch_20newsgroups(
    subset='all',
    remove=('headers', 'footers', 'quotes')
)
documents = newsgroups.data[:500]
```

Step 3: Preprocessing with spaCy

```
#Information_Retrieval_3
#SANJAY_S_241801246
def preprocess_spacy(text):
    doc = nlp(text.lower())
    tokens = [
        token.lemma_
        for token in doc
        if token.is_alpha and not token.is_stop
    ]
    return ' '.join(tokens)
```

Step 4: Apply Preprocessing

```
#Information_Retrieval_3
#SANJAY_S_241801246
df = pd.DataFrame({'Text': documents})
df['clean_text'] = df['Text'].apply(preprocess_spacy)
df['clean_text'].head()
```

	clean_text
0	sure basher pen fan pretty confused lack kind ...
1	brother market high performance video card sup...
2	finally say dream mediterranean new area great...
3	think scsi card dma transfer disk scsi card dm...
4	old jasmine drive use new system understanding...

dtype: object

Step 5: Extract Stop Words

```
#Information_Retrieval_3
#SANJAY_S_241801246
def extract_stop_words(text):
    doc = nlp(text.lower())
    stop_words_in_text = [
        token.lemma_
        for token in doc
        if token.is_stop and token.is_alpha
    ]
    return ' '.join(stop_words_in_text)
df['stop_words'] = df['Text'].apply(extract_stop_words)
df['stop_words'].head()
```

	stop_words
0	I be some of be about the of any of about the ...
1	my be in the for a that with do anyone have on...
2	you what you about that be the will be after s...
3	it the do the not the the can do from any of t...
4	I have an which I can not with my my be that I...

dtype: object

Step 6: POS Tagging

```
#Information_Retrieval_3
#SANJAY_S_241801246
for i, doc in enumerate(df['Text'][:3]):
    print(f"\nDocument {i+1} POS Tags:")
    spacy_doc = nlp(doc)
    print([(token.text, token.pos_) for token in spacy_doc[:10]])
```

```
Document 1 POS Tags:
[('\n\n', 'SPACE'), ('I', 'PRON'), ('am', 'AUX'), ('sure', 'ADJ'), ('some', 'DET'), ('bashers', 'NOUN'), ('of', 'ADP'), ('Pens', 'PROPN'), ('fans', 'NOUN'), ('are', 'AUX')]

Document 2 POS Tags:
[('My', 'PRON'), ('brother', 'NOUN'), ('is', 'AUX'), ('in', 'ADP'), ('the', 'DET'), ('market', 'NOUN'), ('for', 'ADP'), ('a', 'DET'), ('high', 'ADJ'), ('-', 'PUNCT')]

Document 3 POS Tags:
[('\n\n\n\n\t', 'SPACE'), ('Finally', 'ADV'), ('you', 'PRON'), ('said', 'VERB'), ('what', 'PRON'), ('you', 'PRON'), ('dream', 'VERB'), ('about', 'ADP'), ('.', 'PUNCT'), ('Mediterranean', 'PROPN')]
```

Step 7: Vectorization using TF-IDF

+ Code + Text

```
#Information_Retrieval_3
#SANJAY_S_241801246
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(df['clean_text'])
```

Step 8: Define the Query

```
#Information_Retrieval_3
#SANJAY_S_241801246
query = "NASA space launch"
query_clean = preprocess_spacy(query)
query_vec = vectorizer.transform([query_clean])
```

Step 9: Calculate Similarity and Rank Documents

```
#Information_Retrieval_3
#SANJAY_S_241801246
similarities = cosine_similarity(
    query_vec,
    tfidf_matrix
).flatten()
top_indices = similarities.argsort()[-5:][::-1]
```

Step 10: Display Top 5 Results

```
#Information_Retrieval_3
#SANJAY_S_241801246
print("\nTop 5 Relevant Documents:")
for i in top_indices:
    print(df.loc[i, 'Text'][:300])
    print()
```

```
Top 5 Relevant Documents:
I am posting this for a friend without internet access. Please inquire
to the phone number and address listed.
---------------------------------------------------------------------

"Space: Teaching's Newest Frontier"
Sponsored by the Planetary Studies Foundation

The Planetary Studies Foundation is

: >Why not build a inflatable space dock.

: If you're doing large-scale satellite servicing, being able to do it in
: a pressurized hangar makes considerable sense.  The question is whether
: anyone is going to be doing large-scale satellite servicing in the near
: future, to the point of justifyin

"Space Station Redesign Leader Says Cost Goal May Be
Impossible"

Today (4/6) the Washington Post ran an article with the
headline shown above. The article starts with "A leader
of the NASA team in charge of redesigning the planned
space station said yesterday the job is tough and may
be impossible.




Phil> Didn't one of the early jet fighters have these?  I also think
Phil> the germans did some work on these in WWII.

The NACA came up with them before World War II.  NASA is directly
descended from the NACA, with space added in.

You'll notice that I didn't mention sweep wings even though t

As for SF and advertising in space. There is a romantic episode
in Mead's "The Big Ball of Wax" where the lovers are watching
the constellation Pepsi Cola rising over the horizon and noting
the some 'stars' had slipped cause the Teamsters were on strike.

This was the inspiration for my article on
```

RESULT:

The information retrieval system using spaCy and TF-IDF was executed successfully. Text was preprocessed using tokenization, stop-word removal, and lemmatization; part-of-speech tags were displayed for sample documents. The query “NASA space launch” was compared with the document collection using cosine similarity, and the top five relevant documents were displayed